

23070521054- Gauri Gulhane

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# =====
# PRACTICAL 1
# Introduction to Deep Learning Frameworks
# TensorFlow / Keras
# =====

import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import numpy as np

print("TensorFlow Version:", tf.__version__)

# -----
# 1. Create sample dataset
# -----
# XOR problem
X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
], dtype=np.float32)

y = np.array([0, 1, 1, 0], dtype=np.float32)

# -----
# 2. Create Neural Network
# -----
model = keras.Sequential([
    layers.Input(shape=(2,)),
    layers.Dense(8, activation='relu'),
    layers.Dense(4, activation='relu'),
    layers.Dense(1, activation='sigmoid')
])

# Display model architecture
print("\nModel Architecture:")
model.summary()

# -----
# 3. Compile the model
# -----
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# -----
# 4. Train the model
# -----
print("\nTraining Model...")

history = model.fit(
    X,
```

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        y,
        epochs=500,
        verbose=0
    )

    print("Training Completed!")

    # -----
    # 5. Evaluate model
    # -----
    loss, accuracy = model.evaluate(X, y, verbose=0)

    print("\nModel Performance")
    print("-----")
    print("Loss      :", round(loss, 4))
    print("Accuracy :", round(accuracy, 4))

    # -----
    # 6. Make predictions
    # -----
    predictions = model.predict(X, verbose=0)

    print("\nPredictions")
    print("-----")
    print("Input\tActual\tPredicted\tProbability")
    print("-----")

    for i in range(len(X)):
        predicted_class = int(predictions[i][0] >= 0.5)

        print(
            X[i].astype(int),
            "\t",
            int(y[i]),
            "\t",
            predicted_class,
            "\t\t",
            round(float(predictions[i][0]), 4)
        )
```

TensorFlow Version: 2.20.0

Model Architecture:

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 8)	24
dense_1 (Dense)	(None, 4)	36
dense_2 (Dense)	(None, 1)	5

Total params: 65 (260.00 B)

Trainable params: 65 (260.00 B)

Non-trainable params: 0 (0.00 B)

Training Model...

Training Completed!

Model Performance

Loss : 0.2741

Accuracy : 1.0

Predictions

Input	Actual	Predicted	Probability

[0 0]	0	0	0.1496
[0 1]	1	1	0.7444
[1 0]	1	1	0.8185
[1 1]	0	0	0.3553